

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 8: Databases

Subtopic 8.1: Database Concepts

Concept 8.1.1: File-Based vs Relational Database Approach

Total Questions: 5

Mark Schemes begin on Page 4

Question 1 | Source: 9618_w24_qp_11 (Q2.a)

2 A shop repairs electronic devices, for example mobile phones and tablet computers. The shop owner stores the data about the repairs using a file-based approach.

(a) Give **one** limitation of using a file-based approach to store the data **and** explain how a relational database addresses this limitation.

Limitation

Explanation

[3]

Question 2 | Source: 9618_w24_qp_12 (Q6.a)

6 A company uses a relational database to store data about its customers, employees and the individual repair jobs that customers have booked.

(a) Explain the benefits of using a relational database instead of a file-based approach.

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[3]

Question 3 | Source: 9618_w23_qp_11 (Q3.b)

3 A shop manager has designed a relational database to store customer orders.

The database will have the following tables:

CUSTOMER(CustomerID, FirstName, LastName, Town)

SHOP_ORDER(OrderNo, CustomerID, OrderDate)

SUPPLIER(SupplierID, EmailAddress, TelephoneNumber)

ITEM(ItemNumber, SupplierID, Description, Price)

ORDER_ITEM(ItemNumber, OrderNo, Quantity)

(b) Identify **three** advantages of a relational database compared to a file-based approach.

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[3]

Question 4 | Source: 9618_s23_qp_13 (Q4.a)

4 A shop rents cars to customers. The shop uses a relational database to store information about the rentals.

(a) Describe **two** ways in which a relational database addresses the limitations of a file-based approach.

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[4]

Question 5 | Source: 9618_s21_qp_11 (Q7.a)

7 Bobby and Kim are discussing databases.

(a) Bobby tells Kim that a file-based approach is usually better than a relational database.

Explain why Bobby is incorrect.

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[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w24_ms_11 (Q2.a)

2(a)	1 mark for limitation and 2 marks for corresponding explanation e.g. Data redundancy // data duplication • Separate linked tables are used • Data items are stored once so reduces data duplication / data redundancy Data inconsistency / Poor data integrity • Data changed once in one place will automatically update elsewhere • Linked data cannot be entered differently in two tables • Referential integrity can be enforced The data structure used depends on the application • Changes to the data structure are managed by the DBMS • ... and queries are not dependent on the structure of the data • ... and changes to the data do not require programs to be re-written	3
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Mark Scheme for Question 2 | Source: 9618_w24_ms_12 (Q6.a)

6(a)	1 mark each to max 3 e.g. • There is reduced data redundancy // less repeated data • ... because each item of data is only stored once • Data consistency is maintained // Data integrity is improved • ... changes in one table will automatically update in another • ... linked data cannot be entered differently in two tables • Program-data independence is ensured • ... changes to the data do not require programs to be re-written // queries are not dependent on the structure of the data • Complex queries are easier to run • Different views can be provided • so users can only see specific aspects of the database • Multiple concurrent access is possible • ... through record locking	3
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Mark Scheme for Question 3 | Source: 9618_w23_ms_11 (Q3.b)

3(b)	1 mark for each bullet point (max 3) - Reduced data redundancy - Improved data integrity / consistency / referential integrity - Allows for views / improved privacy - Allows for program-data independence - Complex queries can be executed	3
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Mark Scheme for Question 4 | Source: 9618_s23_ms_13 (Q4.a)

4(a)	1 mark for each bullet point. Mark in pairs. Max 2 for each description. - Reduces data redundancy ... because linked tables mean that each data item is stored only once - Reduces program-data dependency ... because the data is separate from the software so changes to the data do not require programs to be re-written - Reduces data inconsistency // improves data integrity ... because by only storing data once it only needs to be updated once // changes in one table will automatically update in another // linked data cannot be entered differently in two tables - Complex queries are easier to run - Can provide different views so users can only see specific aspects of the database	4
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Mark Scheme for Question 5 | Source: 9618_s21_ms_11 (Q7.a)

7(a)	1 mark per bullet point to max 3 - Flat-file has more data redundancy ... because the same data is stored many times // data is stored in different tables which are linked - There is program-data dependence with flat-files ... because any changes to the structure of the data means the programs that access that data have to be re-written - Flat-file has more data inconsistency // worse data integrity ... because duplicated data might be stored differently //...because when data is updated in one place, it is not updated everywhere - It is not easy to perform complex searches /queries ... because a new program has to be written each time - Flat files could have a lack of privacy ... as user views cannot easily be implemented	3
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